

MODELING ELECTORAL BEHAVIOUR WITH PENNSYLVANIA ELECTORAL DATA

SMAC TALK

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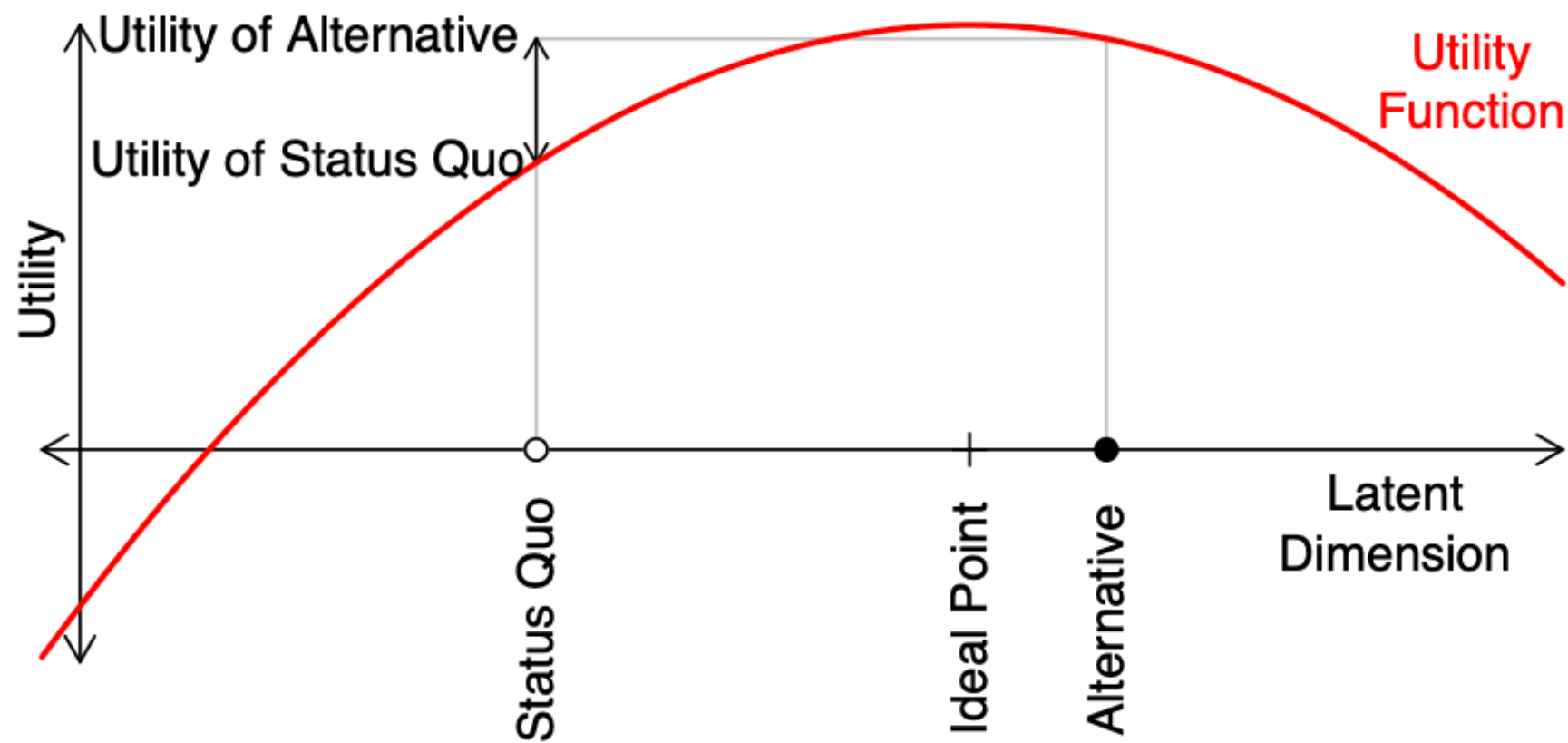
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- ▶ In our project we connected one of the most used models of political science (spatial voting model) with a structural model from economics
- ▶ Then, we proposed a method to estimate the parameters of the model

SPATIAL VOTING MODEL

- ▶ The basic assumption of spatial voting model is that you can put voters and candidates on a same political space, and voters will choose the closest candidate



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- ▶ More in general, each choice structure compatible with a spatial voting model can be described by the following utility function:

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- ▶ Each voter chooses the candidate that maximises the utility
- ▶ Following Merlo and Paula (2017) we know that if we impose the same weighting matrix to each voter, then we are able to identify the distribution of voters' positions

MERLO AND DE PAULA(2017)

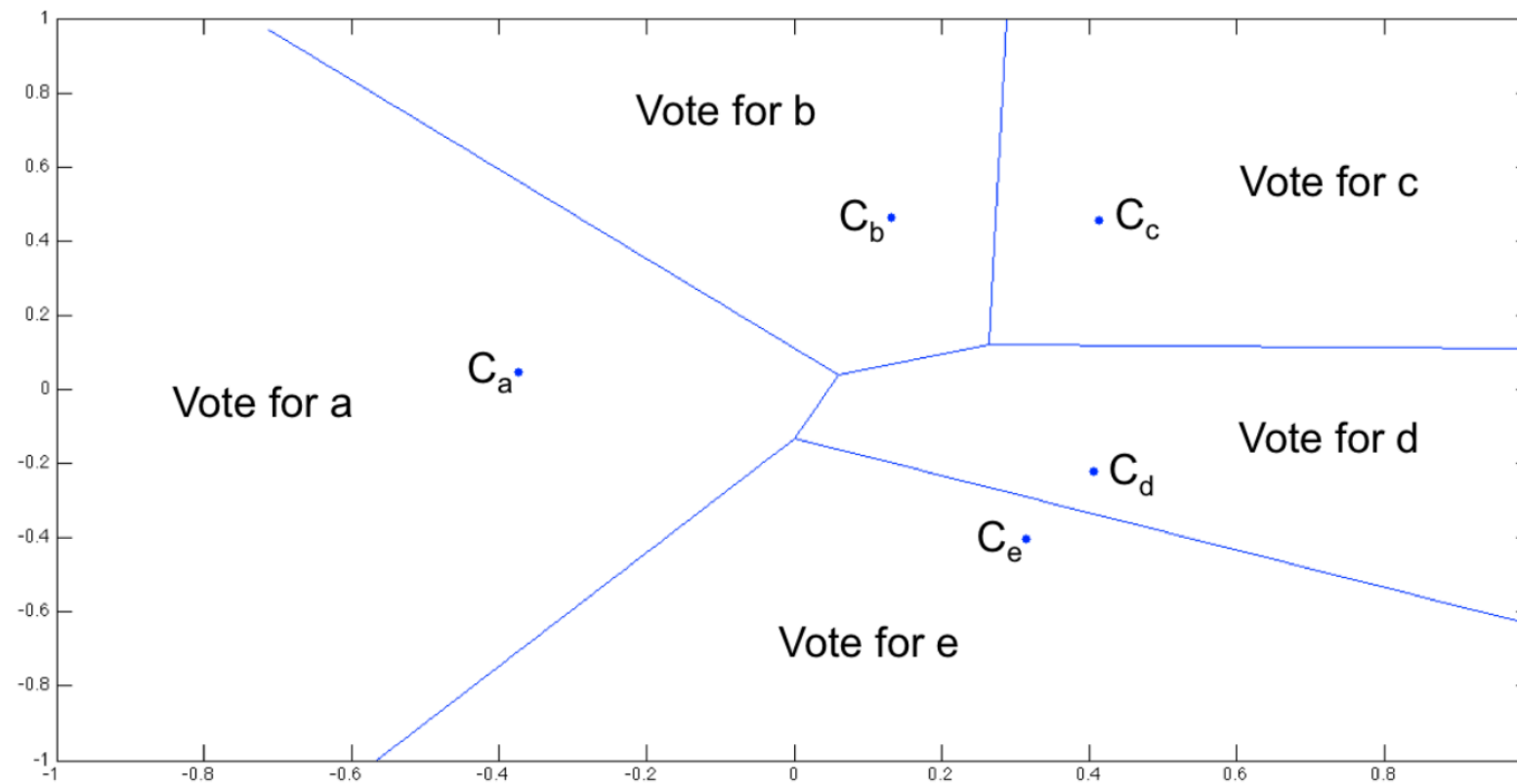


Figure 1: The Voronoi Tesselation for a 5-candidate election in $\mathbb{R}^2$ and $W = I$.

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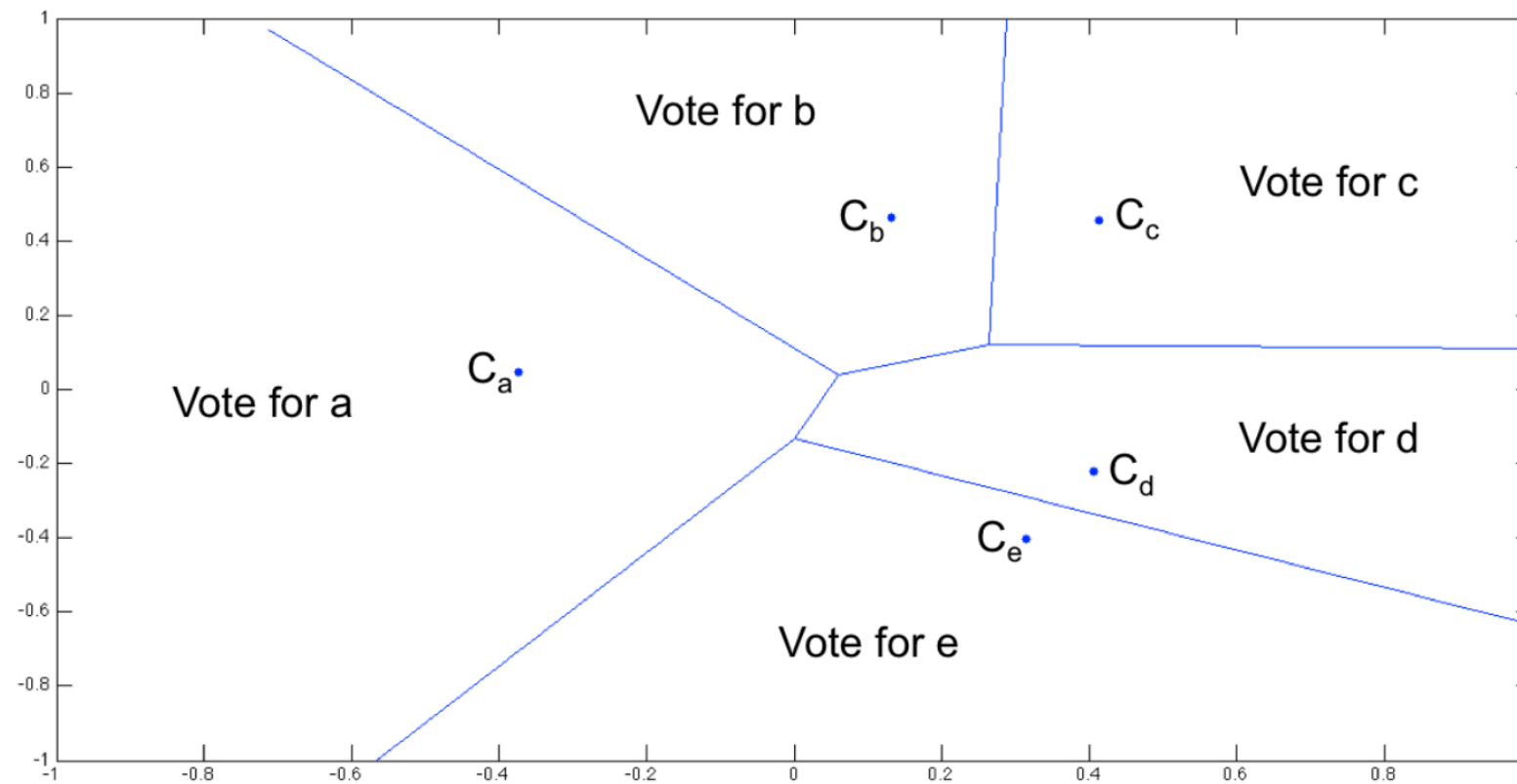


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- If each election is a different tessellation of the political space, then it is a matter of measuring the subspaces

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- ▶ That means that, even if candidates behave strategically, their strategic behaviour is constrained by what they can observe about electors

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- ▶ $\Pr(y_{iD} = 1 \mid \mathbf{v}_i) = \Pr((\epsilon_{iD} - \epsilon_{iR}) \leq (\mathbf{x}_D - \mathbf{x}_R)'\mathbf{B}\mathbf{v}_i)$

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- ▶ However, we can't observe how each voter votes (luckily so!), therefore we need to work with aggregate data: our smallest unit of observation is the electoral precinct

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- ▶ From ALARM dataset (Kenny and McCartan, 2021), we had electoral results and ethnic composition at precinct level (s_{Dg} and part of $\bar{c}_g$)
- ▶ From census data, we took data on educational attainment and reapportioned them at precinct level (remaining part of $\bar{c}_g$)
- ▶ Finally, we looked for data on candidates' ideology: we needed a score both for winning and losing candidates, then we adopted DIME score (Bonica, 2024) ($\mathbf{x}_D, \mathbf{x}_R$)

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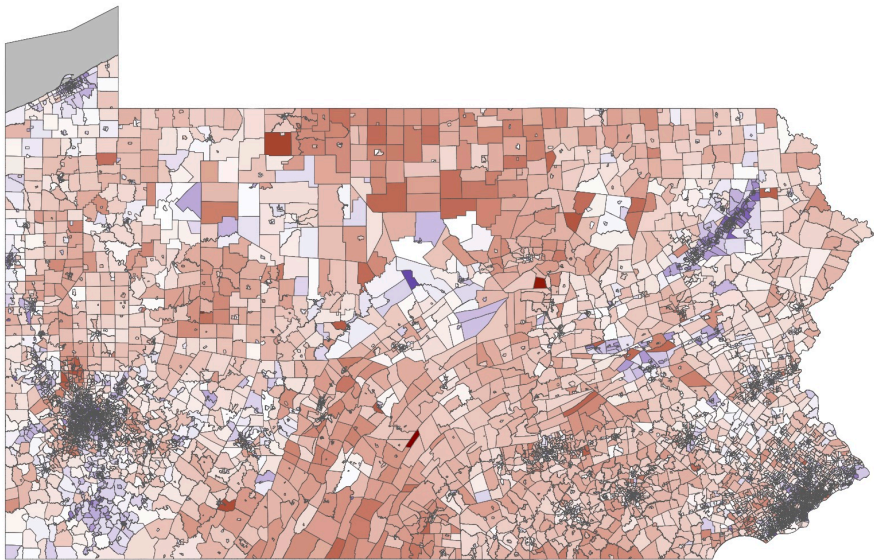
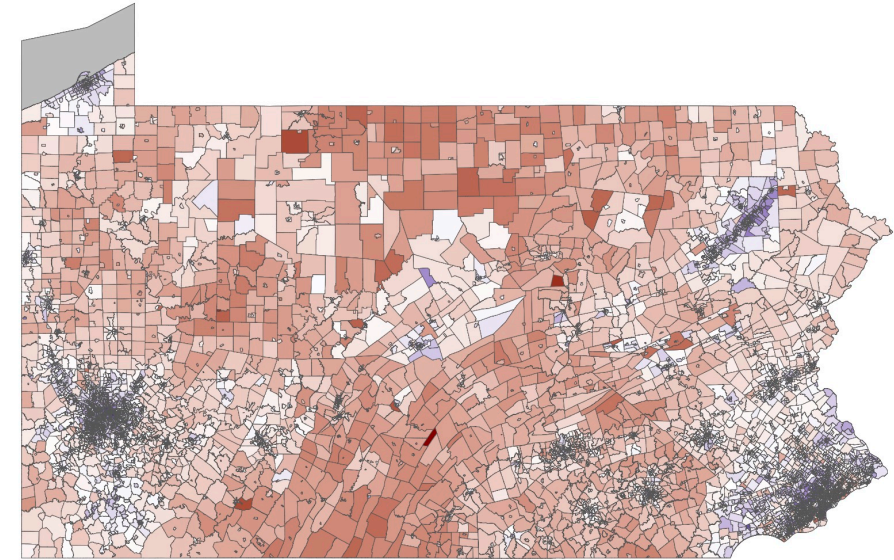
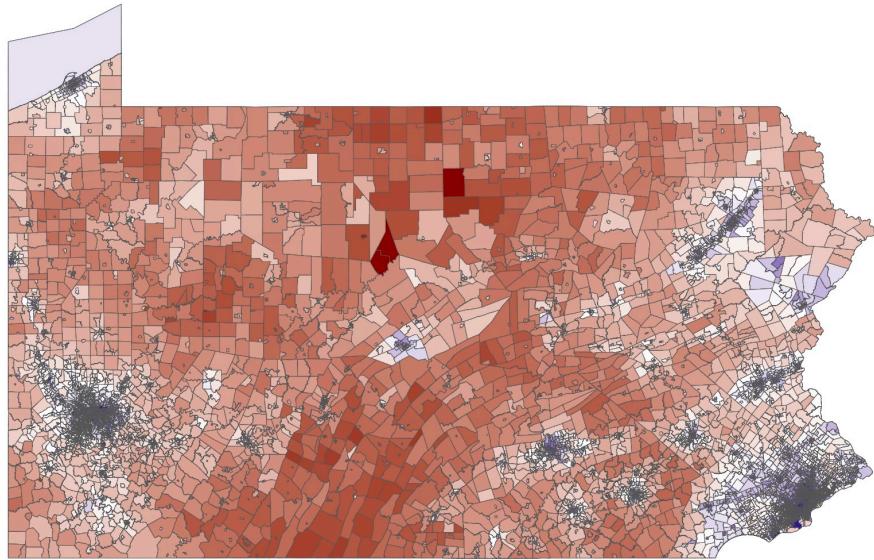
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- ▶ We are now able to estimate relative ideological positions of precincts ($\frac{w\bar{x}_g}{a}$) and the effect of each covariate ($\frac{\tilde{\beta}_{1m}}{a}$)

ESTIMATES OF IDEOLOGICAL POSITIONS WITHIN PRECINCTS



- ▶ top-left: average Democratic share in each precinct in statewide elections from 2016 to 2020
- ▶ top-right: ideological estimate of each precinct accounting for ethnical composition of precincts
- ▶ bottom-left: ideological estimate of each precinct accounting for both ethnical composition and share of people with a college degree

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- ▶ The estimates of both ideological positions of precincts and effects of demographic characteristics are not very interesting: we are mainly decomposing partisanship into ethnic and education features
- ▶ That, however, is quite different from our original scope
- ▶ The main issue there is the monodimensional political space

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ESTIMATING CANDIDATES' POSITIONS IN A MULTIDIMENSIONAL SPACE

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- ▶ We are then interested in placing candidates in a multidimensional space: that also helps in understanding the effect of personal characteristics on voters' choices
- ▶ Various methodologies have been applied to this problem (voting scores, candidates' websites) but there is a lack of public datasets that cover both winners and losers
- ▶ Then, we tried to develop our own

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- ▶ We just adopt a prior on distribution of precinct positions (based on what we estimated before, a Normal distribution seems to suit) and a prior on ideological positions of the candidates on the first dimension (based on DIME)